

1 The highest neural selectivity was identified within the high-gamma frequency band.
2 Consequently, we hypothesized that robust decoding accuracy could be achieved by utilizing
3 bandpass-filtered data between 70-150 Hz. This decoding framework was systematically
4 applied to both species.

5 To map the precise temporal evolution of information processing, we implemented a time-
6 resolved decoding architecture on the MATLAB environment. In this scheme, a classification
7 model is trained at every individual time bin, with the input feature vector defined by the filtered
8 LFP amplitudes across all available channels.

9 However, before delving into the results, it is essential to address the inherent data constraints.
10 The primary challenge in the human dataset is the severe scarcity of valid trials per class. This
11 constraint is absent in the non-human primate dataset, where a highly controlled experimental
12 design yielded a consistent baseline of at least 12 correct trials per class for both the spatial
13 and shape tasks. In contrast, many human SEEG channels lacked sufficient trials across all
14 conditions. Because the feature vector dimension must remain constant throughout the time-
15 resolved decoding pipeline, any channel missing data for even a single class had to be
16 discarded. To balance this channel drop-out against the need for a sufficiently high-
17 dimensional feature space to yield statistically meaningful conclusions, the human dataset
18 was strictly standardized to 3 trials per class, maximizing the number of preserved channels.

19 Given this low trial count, we employed a robust Stratified Leave-One-Per-Class-Out (SLOO)
20 cross-validation scheme. The data were partitioned into k-folds to isolate completely
21 independent training and testing sets, ensuring rigorous validation.

22 We evaluated two distinct channel-inclusion strategies. In the first approach, all surviving
23 channels within a given region (Dorsal or Ventral) were utilized. In the second approach, we
24 integrated an feature-selection step directly into the pipeline to isolate the most informative

channels. Using the mean high-gamma power calculated exclusively during the cue period, a preliminary linear support vector machine (SVM) was trained to differentiate the classes. The channels were then ranked based on the absolute magnitude of their assigned weights, and only the top 50% were selected for final decoding. Crucially, this feature-selection step was nested inside each cross-validation fold using training data alone, strictly preventing any data leakage from the test set.

The core classifier utilized across both approaches was a linear SVM with Ridge regularization, following a similar approach as in previous studies [1,2]. The regularization hyperparameter λ was scaled logically based on the feature selection mode. Because λ controls the penalty imposed on non-informative features, a higher λ was assigned to the "All Channels" approach to aggressively suppress the noise introduced by non-informative channels. Conversely, a lower λ was sufficient for the "Top 50%" approach, where the feature vector had already been filtered for task-relevant information. The interested reader could refer to [3,4] for more information regarding Ridge regularization.

As mentioned, a linear kernel is selected. The reasoning is that in high-dimensional spaces, data points are often already linearly separable. Thus, introducing a complex, non-linear kernel drastically would increase the risk of overfitting on the limited trial noise. Recall that our available training sample size is severely constrained, especially in the human dataset with 3 trials per class. In addition to that, a One-vs-All multiclass strategy was employed, mainly for its efficiency and its physiological alignment.

Moreover, a moving average window (500 ms for NHP and ~488 ms for human datasets) was applied to the bandpass-filtered amplitude envelopes prior to decoding. This smoothing step serves two critical purposes. Firstly, since the high-gamma amplitude envelopes are inherently noisy at microsecond resolutions, temporal smoothing filters out high-frequency trial-to-trial

volatility. In addition to that, it broadens the informative window slightly, allowing the classifier to reliably capture stable, sustained state transitions (such as Cue Onset or Delay maintenance) rather than getting confused by transient, phase-dependent fluctuations. Following the moving average filter, the time series were temporally downsampled by a step size of 10 samples in both datasets (20 ms steps for NHP and ~19.5 ms steps for humans). In addition, a local z-score normalization was applied to each trial. Specifically, each channel's time series was z-scored using the mean and standard deviation calculated from a restricted 500 ms fixation window (0.25-0.75s post-fixation onset). This specific interval was utilized instead of the full fixation period to prevent temporal overlap with either the initialization transients of the trial or the sensory onset of the cue period.

To ensure an unbiased comparison between the two regions, which originally contained different numbers of electrodes, a channel-resampling procedure was implemented. For each region a matched population of channels (e.g. $N = 150$ for NHP; $N = 70$ for humans in the spatial task) was randomly subsampled. To guarantee that the decoding results were robust and independent of any single lucky electrode configuration, this random subsampling process was repeated across 10 independent iterations (permutations). The final solid decoding curves reflect the grand average across all 10 iterations and their respective cross-validation folds. To visualize the baseline trial-to-trial volatility without shrinking the error bounds, the standard error of the mean (SEM) was computed strictly across the cross-validation folds of the first random channel permutation.

Finally, a nonparametric cluster-based permutation test ($p < 0.05$) was implemented to evaluate the statistical significance of the decoding accuracy differences between the two brain regions of interest across time.

The results show a higher classification accuracy in the Dorsal region for both species at least in the spatial task (Fig 1). A spiking in the decoding is noticeable at times of interest and the difference between the two regions is highly significant.

Regarding the spatial task, in the non-human primate dataset, the Dorsal stream (blue) shows a rapid surge in decoding accuracy immediately following cue presentation (1.0 s), peaking at over 50%. While the Ventral stream (orange) also rises above chance baseline (11.1%), its peak is drastically attenuated compared to Dorsal. Nonparametric cluster-based permutation testing confirms that this Dorsal superiority is highly significant (green horizontal bars) and persists uninterrupted across the entire cue presentation and throughout the subsequent periods (second stimulus and response). A highly consistent temporal profile emerges in the human intracranial data. Despite the trial scarcity constraints imposed on the human decoder, the Dorsal stream yields distinct, synchronized decoding spikes at the epochs of the task. Just like the non-human primate profile, the human Dorsal stream shows a robust accuracy peak during the cue period (1.5 s).

On the contrary, during the shape feature working memory task, no meaningful decoding could be achieved in either species. The classification accuracy across both the Dorsal and Ventral streams remains largely clustered at or near either the statistical chance or the fixation baseline, showing no clear information spikes or significant regional divergence during the cue presentation or other epochs.



Monkey



Human

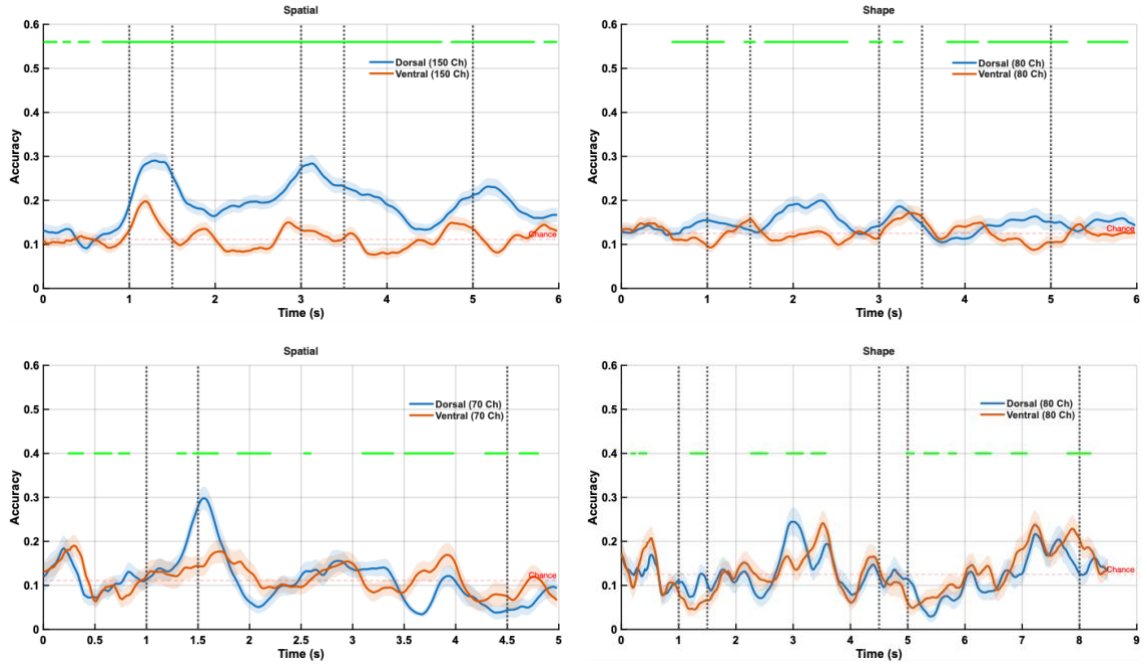


Figure 1. Time-resolved high-gamma decoding accuracy across all available prefrontal channels. The solid lines indicate grand mean classification accuracy averaged across cross-validation folds and channel-resampling iterations (where an identical number of channels was randomly subsampled per region to ensure an unbiased comparison). Shaded areas illustrate the standard error of the mean (SEM) calculated across the cross-validation folds of the initial channel permutation. Horizontal red dashed lines indicate statistical chance level. Green horizontal bars at the top denote temporal clusters where the difference between dorsal and ventral decoding accuracy is highly significant ($p < 0.05$, nonparametric cluster-based permutation test). Vertical black dashed lines indicate task epoch transitions.

Highly consistent results were observed when following the targeted selection of the top 50% most informative channels as discussed earlier (Fig 2). Despite the reduction in feature dimensions, the monkey/human decoder reproduced the exact same synchronized information spikes, with the significance clusters remaining mostly intact.

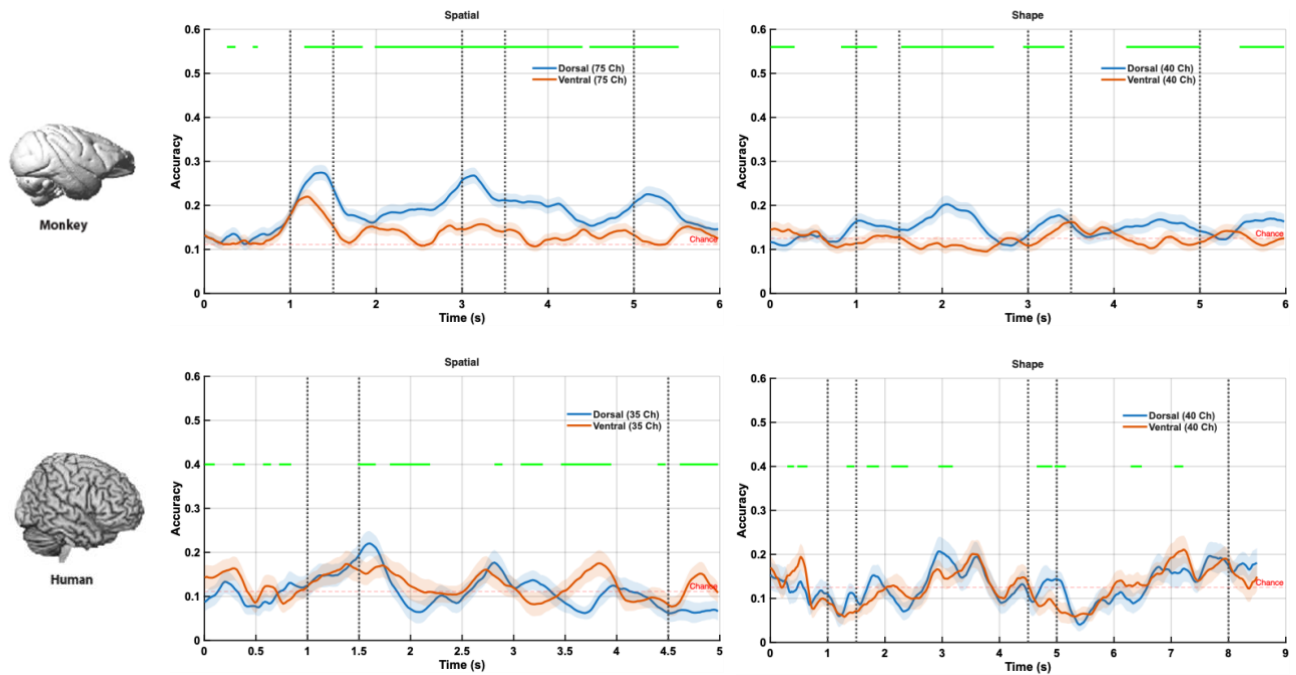


Figure 2. Time-resolved high-gamma decoding accuracy following informative channel selection. The solid lines indicate grand mean classification accuracy averaged across cross-validation folds and channel-resampling iterations (where an identical number of channels was randomly subsampled per region to ensure an unbiased comparison). Shaded areas illustrate the standard error of the mean (SEM) calculated across the cross-validation folds of the initial channel permutation. Horizontal red dashed lines indicate statistical chance level. Green horizontal bars at the top denote temporal clusters where the difference between dorsal and ventral decoding accuracy is highly significant ($p < 0.05$, nonparametric cluster-based permutation test). Vertical black dashed lines indicate task epoch transitions.

Methods

The decoding pipeline was based on a supervised, linear support vector machine (SVM) classifier implemented in MATLAB using the built-in `fitcecoc` function. This function used a One-vs-All multiclass strategy to decode the spatial target conditions. Prior to decoding, the bandpass-filtered high-gamma amplitude envelopes were smoothed using a moving average window (500 ms for NHP and ~488 ms for humans) and temporally downsampled using a step size of 10 samples. This step size resulted in a time-resolved timeline with 20 ms steps for the NHP data and ~19.5 ms steps for the human data. Following the moving average, a local z-score normalization was applied to each trial. Each channel's time series was normalized using

132 the mean and standard deviation calculated strictly from a 500 ms fixation window (0.25-0.75s
133 post-fixation onset) to avoid any artifacts from trial initialization or early cue activity.

134 To ensure an unbiased comparison between the brain regions, an equal number of channels
135 was randomly selected from each region pool without replacement. This random selection
136 process was repeated across 10 independent outer iterations (permutations). For each
137 selection, the data were organized into a 3D matrix of [Trials $\times$ Channels $\times$ Time Bins]. A
138 Stratified Leave-One-Per-Class-Out (SLOO) cross-validation scheme was implemented,
139 consisting of 60 folds for the NHP dataset and 15 folds for the human dataset. The number of
140 folds was chosen based on the available trials per class for each species. The NHP dataset
141 contained 12 trials per class, while the human dataset contained 3 trials per class. Multiplying
142 these baseline trial counts by 5 determined the total number of cross-validation folds for each
143 case. In each cross-validation fold, exactly one trial per class was randomly left out to form the
144 testing set, while the remaining trials were used for training.

145 In the case of channels selection, inside each cross-validation fold, a linear SVM feature
146 selection step was performed using the training data alone to rank channels based on their
147 absolute weight magnitudes during the cue period; decoding was then evaluated using the top
148 50% highest-ranked channels. The absolute magnitude of weights across all binary learners
149 was used for that purpose.

150 The SVM models were trained and tested dynamically at each individual time bin. To show a
151 realistic snapshot of trial-to-trial cross-validation volatility, the shaded standard error of the
152 mean (SEM) was calculated across the cross-validation folds of the first outer channel
153 permutation, while the solid center line represents the grand average across all 10 channel-
154 resampling iterations.

Finally, to determine whether the decoding accuracy differences between the dorsal and ventral regions were statistically significant across time, a nonparametric cluster-based permutation test was performed using 1,000 shuffles. For this test, the full distribution of cross-validation folds across all 10 channel iterations (600 total folds for NHP, 150 total folds for humans) was pooled to maximize statistical power. Significant temporal clusters were identified where the true regional decoding difference surpassed the $p < 0.05$ threshold relative to the permuted null distribution.

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